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Evaluating the Performance of the Suppliers Using Hybrid DEA-OPA Model: A Sustainable Development Perspective

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Abstract
One of the most important activities in any organization is the identification and selection of the right supplier. The responsible organizations determine the performance of their potential suppliers based on the attributes that are aligned with their sustainable development goals. Data regarding these attributes can be qualitative and/or quantitative, and not every supplier selection methodology can handle them simultaneously. Data envelopment analysis (DEA) is an influential methodology for measuring the suppliers' performance, especially when there are more than one input and/or output. However, the original DEA model cannot consider human judgment during evaluation. In this regard, the current study proposes a novel methodology with the aid of the Ordinal Priority Approach (OPA) of multiple attributes decision-making and the DEA. The proposed method, the DEA-OPA model, truly enjoys the advantages of both DEA and OPA models, making it more powerful than the original DEA for performance measurement. The proposed model was compared with the original DEA and OPA then retested through incomplete data when the experts lack enough knowledge about inputs and outputs. Finally, a pilot application has been executed in the paper industry, and comprehensive sensitivity analysis has been performed to illustrate the feasibility of the proposed model. The findings are of significant importance to responsible enterprises and organizational decision-makers.

Keywords Multi-criteria decision analysis · Sustainable development · Data envelopment analysis · Ordinal priority approach · Supply chain management · Performance measurement

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1 Introduction

Since the devastating effects of human activities on the environment are becoming more apparent, the need to strengthen the sustainability approach is mounting (Izadikhah and Farzipoor Saen 2019). Human activities not only generate revenues and create employment opportunities through economic and industrial activities but also produce emissions, cause environmental degradation, and hurt Earth's inhabitants (Deng et al. 2019; Javed et al. 2020). Hence, it is necessary to prevent further degradation of the environment by applying the sustainability-oriented approach in all human activates. Supplier selection is a common problem in human industrial life that concerns every organization on the planet. It has also been recognized as one of the most important supply chain management problems (Tao et al. 2020). For encouraging the suppliers to follow the sustainability approach, the supplier selection process should be performed based on sustainability criteria (Hendiani et al. 2020). In other words, the efficiency of the suppliers should be determined in terms of implementing the sustainable approach. The sustainable supplier selection is the minimum effort that an organization can do to play a positive role in humankind's journey to sustainable development. Different scholars have solved the sustainable supplier selection problem differently using various methods and techniques.

Data envelopment analysis (DEA) is one of the most well-known approaches for gauging supplier performance. DEA employs a nonparametric linear programming model whose usage is significant in real-life conditions to measure the performance of the decision-making units (DMUs). Behind the advantages of the original DEA model, there are some shortcomings as well. The DEA model cannot consider human judgment (qualitative or quantitative), or in other words, unrealistic solutions may be resulted. Therefore, while recognizing that no performance management tool is ideal, a question may arise on how human judgment can be applied while calculating the efficiency of the suppliers based on the opinion of a group of experts. The Ordinal Priority Approach (OPA) is a novel multiple attribute decision-making (MADM) method that was proposed recently (Ataei et al. 2020). It enjoys many benefits compared with other existing MADM methods, such as handling incomplete data, integrating the three dimensions of the decision-making process, which can simultaneously calculate the weights of experts (group decision making), attributes, and alternatives. On the other hand, this method does not require normalization methods, pairwise comparison, aggregation operators, etc. Hence, it can be an excellent candidate to combine with the DEA model for considering human judgment in performance measurement.

Combining the DEA and the OPA models manifests a tradeoff between human judgment and the actual data about the system in real-life that can provide more consistent results compared with when only actual data of the system is considered. Therefore, the contribution of the current study is manifold and can be presented as follows:

The DEA-OPA model, which can determine the performance of the DMUs based on both human judgment and actual input/output data, is proposed.

Calculation of the performance of the DMUs when there are missing values in the input data due to lack of knowledge of the experts about the input/output data.

A comprehensive sensitivity analysis is provided, which illustrates the priority of the experts and even the priority of the human judgment (the OPA model) and actual data of the system (the DEA model). It is a tradeoff that could be performed based on the decision-maker's opinion.

The rest of the study has been organized as follows: first, a literature review has been performed on sustainable supplier selection and the DEA model. Next, the original OPA and DEA models have been reviewed by the authors. After that, the DEA-OPA model has been proposed with the complete procedure and steps. Then, an illustrative example has been presented to show the feasibility of the proposed model and also has been compared its results with the original DEA model. Finally, a pilot application has been executed to illustrate the application of the proposed approach in real-life. Moreover, a comprehensive sensitivity analysis has been performed to demonstrate the flexibility of the proposed approach in different situations.

2 Literature Review

Sustainable supplier selection and evaluating the performance of the suppliers is a popular topic in recent decades. There are different approaches to measure the efficiency of the suppliers, yet employing the DEA is common among many of the studies due to its simplicity and advantages. Dobos and Vörösmarty (2014) considered two types of criteria for assessing the suppliers include environmental and traditional criteria. They employed a common weight analysis (CWA) model in the DEA approach to managing the supplier selection process based on managerial and green factors. Azadi et al. (2015) combined both DEA and enhanced Russell measure model to develop a new method for evaluating the efficiency of the suppliers based on sustainability indexes. They believed that there is uncertainty during a sustainable supplier selection process, and the data of the inputs/outputs cannot be exact in some situations. Therefore, they utilized a fuzzy set to consider the uncertainties in the input data. Zhou et al. (2016) stated that fuzzy DEA is limited to consider different factors of uncertainty in the input and outputs. To overcome this barrier, they proposed a type-2 fuzzy multi-objective DEA model. They believed that the proposed multi-objective DEA model is suitable for balancing three important sustainability dimensions: environmental, social, and economic. Moreover, they demonstrated the capability of the model by executing a case study under uncertainty in a real-world situation. Ji et al. (2016) implemented the sustainability approach to solve the eco-design issue in the transportation industry. They proposed a tractable algorithm with the aid of the DEA, which was capable of finding the Pareto optimal solution even when a large amount of data was available. They illustrated that after employing their algorithm, it could lead to a reduction in pollution and resource consumption in real-life situations. Yousefi et al. (2017) proposed a new hybrid approach utilizing goal programming and the DEA approach to measure efficiency in the sustainable supply chain management context. Moreover, they used fuzzy possibilistic to deal with existing uncertainties in input and output data. Cheng et al. (2017) proposed

a hybrid approach employing the DEA and Adaboost model to select the most suitable supplier when we face a large number of data in the decision-making process. They declared that this hybrid DEA-Adaboost model could decrease the time and computational costs, which can improve the supplier selection process. Jauhar et al. (2017) utilized the Differential Evolution (DE) algorithm and DEA to determine sustainable supply chain efficiency. They executed a case study in an Indian higher educational institute housing nineteen departments. They considered environmental factors to measure the efficiency of the suppliers, which is one of the most important aspects of the sustainability approach. In another research, Jauhar and Pant (2017) argued that to overcoming the drawbacks of the DEA method, it could be integrated with DE and Multi-Objective Differential Evolution (MODE) algorithms. They have implemented their method through a case study in the automobile industry to evaluate its performance. They believed that a combination of DE, MODE, and DEA could present more realistic results compared with the original DEA model. Shabanpour et al. (2017) tried to rank the suppliers based on sustainability criteria utilizing DEA and goal programming. They mentioned that the inputs and outputs data are not precise in most cases; therefore, a robust Charnes-Cooper-Rhodes (CCR) model can be employed in the first-level goals. Suganthi (2018) employed a fuzzy AHP-DEA approach to evaluate sectoral investments towards sustainable development. He emphasized the environmental aspect of the sustainability approach with the aid of two hybrid models includes fuzzy AHP-VIKOR and AHP-DEA. The AHP-VIKOR model was utilized to rank the sustainability criteria and AHP-DEA to investigate that national investment is along with sustainable development or not. Izadikhah and Saen (2018) proposed a chance-constrained two-stage DEA model to evaluate the sustainability of supply chains. They mentioned that their model could deal with stochastic data while there are undesirable factors. Gupta et al. (2018) utilized the AHP-DEA model to determine the efficiency of the vehicles in coal mining to achieve sustainable transportation. Torres-Ruiz and Ravindran (2019) proposed a framework including three stages for sustainable supplier management. They employed DEA with interval data and goal programming method for this matter. Alikhani et al. (2019) believed that risk and sustainability criteria should be considered simultaneously in the supplier selection problem. They proposed a new extension for the DEA model, which could consider desirable and undesirable factors to determine the efficiency of the suppliers. It should be noted that they employed interval type-2 fuzzy sets to collect the input data for the inputs and outputs. Dobos and Vörösmarty (2019) considered the green criteria as output and management variables as input in the supplier selection problem using the DEA model. They declared that the proposed framework could handle inventory-related costs in the supplier selection process. Rashidi and Cullinane (2019) presented a comprehensive comparison between fuzzy TOPSIS and fuzzy DEA in a sustainable supplier selection problem. Based on their results, the fuzzy TOPSIS method is less complicated and sensitive compared with the fuzzy DEA. They executed a real-world case study in Sweden, where both methods revealed the same best supplier. Vörösmarty and Dobos (2020) provided a review on sustainable supplier selection problems with the aid of the DEA model from 2009 to 2018. They stated that there are just a few practical-oriented research items in this context among many conducted studies.

Davoudabadi et al. (2020) believed that flexibility plays an important role in supplier selection problems. Therefore, they investigated the resilient supplier selection problem using an integrated approach that includes principal component analysis, entropy, and DEA. Tavassoli et al. (2020) stated that inputs and outputs data are not always deterministic in the real-world. Hence, they presented a stochastic-fuzzy DEA model that could be used in both certain and uncertain conditions. They mentioned that the α-cut method could be employed to solve crisp problems. Based on the aforementioned studies, there are a few research items that employed the conventional DEA model individually. Scholars have tried to provide hybrid approaches to improve the ability of the DEA under various conditions. The supplier selection process is a sensitive process, and managers should pay special attention to it. The DEA model can calculate the efficiency of the suppliers based on their inputs and outputs data, but it is not enough for making such an important decision.

In Table 1, the proposed approach is compared with the other methods which were used for supplier selection and evaluation. Since the current study combines the DEA and an MCDM method, it should be compared with the literatures with the same attitude.

Based on Table 1, the current study aims to provide a hybrid DEA-OPA method that can balance real-world data and human judgment to make a better decision with less risk. Especially, group decision-making can improve the decision-making process because of considering the opinions of several experts simultaneously. However, the averaging operators are sensitive to outliers, and their results may not be reliable, which is not recommended for experts 'opinions aggregation in group decision-making problems. Moreover, the experts do not have the same knowledge and experience, and their opinions should not be considered equally. The DEA-OPA model is a useful approach for supplier selection because it can handle problems involving multiple experts and incompleteness in data as well. In fact, when the experts do not have enough knowledge about inputs and/or outputs, they can leave it empty. Also, the DEA-OPA model can consider the priority of the experts based on certain distinguishing factors (e.g., working experience, skills, academic degree, etc.) in the supplier selection problem.

Choosing suitable attributes for supplier selection and evaluation plays a vital role in the final decision. From the perspective of sustainable development, the scholars used various attributes from different dimensions of the sustainability approach. In the dimension of the economic, quality, cost, technology capabilities, and delivery time are recognized as the most common attributes among the scholars (Govindan et al. 2015; Zimmer et al. 2016; Rashidi et al. 2020). In the dimension of the environment, environmental management system, pollution control, and recycling are identified as the most popular attributes among scholars (Govindan et al. 2015; Zimmer et al. 2016; Rashidi et al. 2020). In the dimension of the Social sustainability, work safety, labor health, employment relationship, and stakeholder involvement are recognized as the most critical attributes by scholars (Govindan et al. 2015; Rashidi et al. 2020). In the case of data availability, these attributes can be helpful for supplier selection and evaluation problems under the sustainability approach. However, other sustainable attributes can be incorporated based on the situation and availability of the data.

Table 1 A comparison between the proposed model with other models for supplier selection and evaluation

Literature	Core method	Incompleteness in experts' opinions is considered?	Integarted multi-objective linear model for the DEA and MCDM method?	Considered preference of DEA over the MCDM method, or vice versa?	Considered a group of experts?	Used averaging operators for experts' opinions aggregation?	Considered priority among experts? (experts are not equally important)
Seydel (2005)	DEA-SMART	No	No	No	No	No	No
Sevkli et al. (2007)	DEA-AHP	No	No	No	Yes	Yes	No
Kuo et al. (2010)	DEA-ANP-ANN	No	No	No	Yes	Yes	No
Zeydan et al. (2011)	DEA-AHP-TOPSIS	No	No	No	Yes	Yes	No
Kuo and Lin (2012)	DEA-ANP	No	No	No	Yes	Yes	No
Sayyah Markabi and Sabbagh (2013)	DEA-GRA	No	No	No	No	No	No
Shafiee et al. (2014)	DEA-DEMATEL	No	No	No	Yes	Yes	No
Shakouri et al. (2014)	DEA-SAW	No	No	No	Yes	Yes	No
Rouyendegh et al. (2020)	DEA-TOPSIS	No	No	No	Yes	Yes	No
Karami et al. (2021)	DEA-PCA-VIKOR	No	No	No	Yes	Yes	No
The current study	DEA-OPA	Yes	Yes	Yes	Yes	No	Yes

AHP Analytic hierarchy process; *ANN* Artificial neural network; *ANP* Analytic network process; *DEA* Data envelopment analysis; *DEMATEL* Decision making trial and evaluation laboratory; *PCA* Principal component analysis; *SAW* Simple additive weighting; *SMART* Simple multi-attribute rating technique; *TOPSIS* Technique for order of preference by similarity to ideal solution; *VIKOR* VlseKriterijumska optimizacija i kompromisno resenje

3 Ordinal Priority Approach (OPA)

The OPA is one of the modern methods in multiple attributes decision-making. This method was developed by Ataei et al. (2020) and can solve different types of MADM problems, including multiple-expert problems and/or problems with missing values in input data, etc. Afterward, the Robust OPA (OPA-R), Grey OPA (OPA-G) and Fuzzy OPA (OPA-F) were proposed by the scholars to employ the OPA under various types of uncertainty (Mahmoudi et al. 2021b, 2021c, 2022). Some scholars employed the OPA for the project selection problem (Mahmoudi et al. 2021a) and evaluating the barriers to implementing distributed ledger technology (Sadeghi et al. 2021). The reasons for choosing the OPA can be summarized as follows:

1. The OPA can handle group decision-making without using aggregation operators. It should be noted that aggregating operators do not have a good performance when we face outliers in data.
2. The OPA can handle the incomplete input data, thus allowing the experts with insufficient knowledge to leave the answer box empty.
3. The OPA does not need normalization methods. Normalization methods are known to influence final results. Also, the OPA does not require complex input data such as pairwise comparison matrix, which can complicate the problems.

In this section, the steps of the OPA have been tailored for combining with the DEA. First, the sets, parameters, and variables are defined as follows:

Sets	
I	Set of experts $\forall i \in I$
J	Set of attributes$\forall j \in$ J
Indexes	
i	Index of the experts $(1, \ldots, p)$
j	Index of preference of the attributes $(1, \ldots, e+f)$
Variables	
Z	Objective function
$w_{ij}{}^{r_j}$	Weight (importance) of j th attribute based expert's opinion i at rank r_j
Parameters	
r_i	The rank of expert i
r_j	The rank of attribute j
e	The number of input attributes
f	The number of output attributes

Then, the steps of the OPA method can be presented as follows:

Step 1: Specify the experts and sort them based on factors like working experience or academic degree, etc.

Step 2: Determining the inputs and outputs attributes and ranking them by each expert.
Step 3: Proposing and solving Model (1) using the data from steps 1 and 2.

$$
\begin{aligned}
&Max\, Z \\
&\text{S.t}: \\
&Z \le r_i\left(r_j\left(w_{ij}^{r_j} - w_{ij}^{r_j+1}\right)\right) && \forall i \text{ and } j \\
&Z \le r_i r_{e+f} w_{ij}^{r_{e+f}} && \forall i \text{ and } j \\
&\sum_{i=1}^{p}\sum_{j=1}^{e+f} w_{ij} = 1 \\
&w_{ij} \ge 0 && \forall i \text{ and } j \\
&\text{where Z: Unrestricted in sign}
\end{aligned}
\tag{1}
$$

Step 4: After solving Model (1), the weights of attributes can be determined by employing Eq. (2).

$$
w_j = \sum_{i=1}^{p} w_{ij} \quad \forall j \tag{2}
$$

As can be seen from the steps of the OPA, this method is straightforward to use, which needs simple input data to obtain the weights of attributes and experts. Since OPA does not require a pairwise comparison, there is not an inconsistency ratio in the model. When the expert does not have sufficient knowledge about one criterion, it is unnecessary to opine it. This is an imposing advantage compare to other MADM methods.

4 Data Envelopment Analysis (DEA)

DEA is one of the most popular methods to measure the efficiency of the decision-making units (DMUs) when there are more than one input and output. Inspired by the ideas of Farrell (1957), DEA came into existence while employing a linear programming approach. Afterward, Charnes et al. (1978) introduced it as a method for measuring the efficiency of the decision-making units. There are different types of DEA models, such as Charnes Cooper Rhodes (CCR) and Banker, Charnes, Cooper (BCC) (Charnes et al. 1978; Banker et al. 1984). However, the CCR model is the most popular, while many scholars have employed it in recent decades. Also, it

enjoys a high level of simplicity. Therefore, the current study focused on the CCR model. First, the variables and parameters are as follows:

Sets	
K	Set of DMUs $\forall k \in K$
J	Set of inputs and outputs $\forall j \in J$
Indexes	
k	Index of the DMUs $(1, \ldots, m)$
j	Index of the inputs $(1, \ldots, e)$ and outputs $(e+1, \ldots, e+f)$
Variables	
θ^{CCR}	Objective function
w_j	The weight of input/output j
Parameters	
x_{jk}	The value of k th DMU at input j
y_{jk}	The value of k th DMU at output j
x_{j0}	The value of functional DMU at input j
y_{j0}	The value of functional DMU at output j
e	The number of inputs
f	The number of outputs

Considering the variables and parameters above, the CCR-DEA model has been presented in Eq. (3).

$$\begin{aligned} & CCR: \theta_0^{CCR} = Max \sum_{j=e+1}^{e+f} w_j y_{j0} \\ & s.t: \\ & \sum_{j=e+1}^{e+f} w_j y_{jk} - \sum_{j=1}^{e} w_j x_{jk} \leq 0 \quad k = 1, \ldots, m \\ & \sum_{j=1}^{e} w_j x_{j0} = 1, \\ & w_j \geq 0 \quad j = 1, \ldots, e+f \end{aligned} \tag{3}$$

After solving Model (3), the values of w_j will be obtained, which can contribute to calculating the efficiency of the DMU_0. However, Model (3) is inappropriate for combining with the OPA model due to constraint $\sum_{j=1}^{e} w_j x_{j0} = 1$. Therefore, we need another form of the CCR model, which can obtain the same efficiency as Model (3). Here, two lemmas and one theorem are reviewed, which are necessary for the current study (Cooper et al. 2004; Zohrehbandian et al. 2010).

Lemma 1 *Assume that θ_k^* is the optimal solution of Model* (3). *Therefore, $(\hat{x}_k, \hat{y}_k) = (\theta_k^* x_k, y_k)$ can be called the projection of DMU_k on the efficient frontier and DMU_j is efficient.*

Lemma 2 *If the coefficients in Model* (3) *are positive* ($w_j \in R^{e+f} \times R^e$ *related to gradient vector of a supporting hyperplane*) *and* $\sum_{j=e+1}^{e+f} w_j y_{jk} - \sum_{j=1}^{e} w_j x_{jk} = 0$, DMU_k *is efficient*.

Zohrehbandian et al. (2010) proposed Model (4) for CCR-DEA, while they affirmed that its obtained efficiency is equivalent to Model (3).

$$\begin{aligned}
&Max \sum_{j=e+1}^{e+f} w_j \hat{y}_{j0} - \sum_{j=1}^{e} w_j \hat{x}_{j0} \\
&s.t: \\
&\sum_{j=e+1}^{e+f} w_j \hat{y}_{jk} - \sum_{j=1}^{e} w_j \hat{x}_{jk} \le 0 \; k = 1, \dots, m \\
&\sum_{j=1}^{e} w_j + \sum_{j=e+1}^{e+f} w_j = 1 \\
&w_j \ge 0 \quad j = 1, \dots, e+f
\end{aligned} \tag{4}$$

Theorem 1 *Where* w_j^* *is the optimal solution of the Model* (4), *we have* $\sum_{j=e+1}^{e+f} w_j^* y_{jk} / \sum_{j=1}^{e} w_j^* x_{jk} = \theta_k^*$.

Proof Production possibility set of $\left(\hat{x}_k, \hat{y}_k\right) = (\theta_k^* x_k, y_k)$ $k = 1, \dots, m$ is similar to the original production possibility set. Moreover, due to the fact that $(\theta_k^* x_k, y_k)$ is the input-based projection of DMU_k on the efficient frontier, it is efficient (see Lemmas 1 and 2). Therefore, the result is $\sum_{j=e+1}^{e+f} w_j^* y_{jk} / \sum_{j=1}^{e} w_j^* x_{jk} = \theta_k^*$.

Based on the aforementioned information, after replacing $\left(\hat{x}_k, \hat{y}_k\right) = (\theta_k^* x_k, y_k)$ in Model (4), Model (5) was resulted.

$$\begin{aligned}
&Max \sum_{j=e+1}^{e+f} w_j y_{j0} - \theta_0^{CCR} \sum_{j=1}^{e} w_j x_{j0} \\
&s.t: \\
&\sum_{j=e+1}^{e+f} w_j y_{jk} - \theta_k^{CCR} \sum_{j=1}^{e} w_j x_{jk} \le 0 \; k = 1, \dots, m \\
&\sum_{j=1}^{e} w_j + \sum_{j=e+1}^{e+f} w_j = 1 \\
&w_j \ge 0 \quad j = 1, \dots, e+f
\end{aligned} \tag{5}$$

The current study aims to employ Model (5) to combine it with the OPA model. Also, the same model was employed to combine the best worst method with the DEA (Omrani et al., 2020). Model (5) contains the following properties:

1. Model (5) can obtain the normalized weights, which the sum of the weights is equal to 1.

2. If we calculate the efficiency using the obtained weights through Model (5), its efficiency is equivalent to the CCR model.
3. When the DMU is efficient, the value of the objective function is equal to zero because improving the inputs and outputs is impossible.

In the next section, we will propose the DEA-OPA Model with the aid of Model (5).

5 The Proposed DEA-OPA Model

In order to consider the expert's opinion while measuring the efficiency of the DMUs in the DEA method, it is essential to make some improvements in the original DEA model. The OPA is a novel and robust method in the MADM field, which can solve various types of MADM problems using simple steps. Therefore, the current study attempts to improve the performance of the original DEA model by utilizing the OPA approach. Considering Models (1) and (5), multiple objectives model (6) can be proposed to make a tradeoff between DEA and OPA models.

$$
\begin{aligned}
&\max f_k = \sum_{j=e+1}^{e+f}\sum_{i=1}^{p} w_{ij}y_{jo} - \theta_0^{CCR}\sum_{j=1}^{e}\sum_{i=1}^{p} w_{ij}x_{jo} \quad k = 1,\ldots,m \\
&\max f_{m+1} = Z \\
&s.t: \\
&\sum_{j=e+1}^{e+f}\sum_{i=1}^{p} w_{ij}y_{jk} - \theta_k^{CCR}\sum_{j=1}^{e}\sum_{i=1}^{p} w_{ij}x_{jk} \le 0 \qquad k = 1,\ldots,m \\
&\sum_{j=1}^{e+f}\sum_{i=1}^{p} w_{ij} = 1 \\
&Z \le r_i\left(r_j\left(w_{ij}^{r_j} - w_{ij}^{r_j+1}\right)\right) \qquad \forall i \text{ and } j \\
&Z \le r_i r_{e+f} w_{ij}^{r_{e+f}} \qquad \forall i \text{ and } j \\
&w_{ij} \ge 0 \qquad \forall i \text{ and } j \\
&\text{where Z: Unrestricted in sign}
\end{aligned}
\tag{6}
$$

Here, it should be noted that the first objective function and the first constraint come from the DEA model. On the other hand, the second objective function, third and fourth constraints, come from the OPA model, which can contribute to the DEA model for considering the experts' opinions. It is worth mentioning that the second constraint is shared between both DEA and OPA models. Since Model (6) is a multiple objective linear programming, it is essential to select a suitable approach to solve it. In this study, the weighted Min–Max method has been employed because it can consider the specific weight for each objective function (Marler and Arora 2004). It is more flexible than the Min–Max method for decision-makers because there is an option to specify which model should play a more important role in the results. Since the scale of DEA objectives is different from the OPA objective, the objective functions should be transformed to non-dimensional objective functions. Equation (7) can be employed to transform the objective functions (Makler and Arora 2005).

$$f_k^{\text{Trans}} = \frac{f_k^* - f_k(x)}{f_k^* - f_k^{\min}},\ k = 1, \ldots, m+1 \tag{7}$$

Suppose that S_{DEA} and S_{OPA} are the weights of the DEA and OPA objectives functions, respectively, while ($S_{DEA} + S_{OPA} = 1$). Therefore, Model (6) can be converted to Min–Max Model (8).

$$\begin{aligned}
&\text{Min Max}\left\{\left(\frac{S_{DEA}}{m}\right)\left[\frac{f_k^* - (\sum_{j=e+1}^{e+f}\sum_{i=1}^{p} w_{ij}y_{jo} - \theta_0^{CCR}\sum_{j=1}^{e}\sum_{i=1}^{p} w_{ij}x_{jo})}{f_k^* - f_k^{\min}}\right], S_{OPA}\left[\frac{f_{m+1}^* - Z}{f_{m+1}^* - f_{m+1}^{\min}}\right]\right\}\\
&k = 1, \ldots, m\\
&s.t:\\
&\sum_{j=e+1}^{e+f}\sum_{i=1}^{p} w_{ij}y_{jk} - \theta_k^{CCR}\sum_{j=1}^{e}\sum_{i=1}^{p} w_{ij}x_{jk} \le 0 && k = 1, \ldots, m\\
&\sum_{j=1}^{e+f}\sum_{i=1}^{p} w_{ij} = 1\\
&Z \le r_i\left(r_j\left(w_{ij}^{r_j} - w_{ij}^{r_j+1}\right)\right) && \forall i \text{ and } j\\
&Z \le r_i r_{e+f} w_{ij}^{r_{e+f}} && \forall i \text{ and } j\\
&w_{ij} \ge 0 && \forall i \text{ and } j\\
&\text{where Z: Unrestricted in sign}
\end{aligned} \tag{8}$$

After utilizing the weighted Min–Max method to solve multiple objective Model (8), it can be written as Model (9), which is called the DEA-OPA model. This model is a single objective; therefore, it can be solved using conventional algorithms in the linear programming context.

$$\begin{aligned}
&\min \lambda\\
&s.t:\\
&\left(S_{DEA}/m\right)\left[\frac{f_k^* - (\sum_{j=e+1}^{e+f}\sum_{i=1}^{p} w_{ij}y_{jo} - \theta_0^{CCR}\sum_{j=1}^{e}\sum_{i=1}^{p} w_{ij}x_{jo})}{f_k^* - f_k^{\min}}\right] - \lambda \le 0 \quad k = 1, \ldots, m\\
&S_{OPA}\left[\frac{f_{m+1}^* - Z}{f_{m+1}^* - f_{m+1}^{\min}}\right] - \lambda \le 0\\
&\sum_{j=e+1}^{e+f}\sum_{i=1}^{p} w_{ij}y_{jk} - \theta_k^{CCR}\sum_{j=1}^{e}\sum_{i=1}^{p} w_{ij}x_{jk} \le 0 && k = 1, \ldots, m\\
&\sum_{j=1}^{e+f}\sum_{i=1}^{p} w_{ij} = 1\\
&Z \le r_i\left(r_j\left(w_{ij}^{r_j} - w_{ij}^{r_j+1}\right)\right) && \forall i \text{ and } j\\
&Z \le r_i r_{e+f} w_{ij}^{r_{e+f}} && \forall i \text{ and } j\\
&w_{ij} \ge 0 && \forall i \text{ and } j\\
&\text{where Z: Unrestricted in sign}
\end{aligned} \tag{9}$$

After solving the proposed DEA-OPA model and finding the optimal values of ${w_{ij}}^*$, Eq. (10) can be employed to determine the efficiency of the DMU k.

$$\theta_k^{DEA-OPA} = \frac{\sum_{j=e+1}^{e+f} \sum_{i=1}^{p} w_{ij}^{DEA-OPA} y_{jk}}{\sum_{j=1}^{e} \sum_{i=1}^{p} w_{ij}^{DEA-OPA} x_{jk}} \tag{10}$$

Based on the information mentioned above, the DEA-OPA model is genuinely comprehensive, while it can calculate the weight of inputs and outputs using experts' opinions as well as quantitative data about the suppliers. It is interesting to mention that DEA-OPA can support missing values as well. When the expert lacks enough knowledge about one output and/or input, there is no need to give an opinion on it. In this case, the constraint related to this input and/or output should be removed from the OPA constraints. This issue has been addressed in the numerical example section carefully. Also, for more details, Ataei et al. (2020) can be referred.

6 The Steps of the Proposed Framework

In this section, we intend to explain the steps of the DEA-OPA model. The steps of the proposed model have been summarized in Fig. 1.

Based on Fig. 1, the steps of the proposed framework have been explained as follows:

Step 1: Identifying the suppliers ($k = 1, ..., m$) and specifying the sustainable inputs (x_{jk}, $j = 1, \dots, e$) and outputs ($y_{jk}, j = e + 1, \dots, e + f$), which are essential for the company. In this step, the values of inputs and outputs should be collected from the system for each supplier. Also, Model (3) should be formed and solved for each supplier. Therefore, θ_k^{CCR} ($k = 1, ..., m$) for each supplier is calculated.
Step 2: Identifying the potential experts in the company ($i = 1, \dots, p$) and ranking them based on a distinguishing factor like working experience, academic degree, etc.(a distinguishing factor $r_i, i = 1, \dots, p$). After that, the data should be collected from the experts regarding the inputs (r_j, $j = 1, \dots, e$) and outputs (r_j, $j = e + 1, \dots, e + f$) attributes. Based on the obtained results in Step 1 and data collected in the current step, Model (6) should be formed and solved by each objective function separately in two states $f_k^{min}, k = 1, \dots, m + 1$ and $f_k^{max}, k = 1, \dots, m + 1$ where $f_k^* = Maximum_x \{f_k(x), k = 1, \dots, m + 1 | x \epsilon X\}$ in Model (6).
Step 3: Determining the values of S_{DEA} and S_{OPA}, the total weight of the objective functions of the DEA and the weight of the objective function of the OPA, respectively. Here, the weights of the objective functions can be estimated through OPA or any other suitable method. Considering the obtained results in Step 2, Model (9) should be formed and solved. Finally, $w_{ij}{}^*, i = 1, \dots, p, j = 1, \dots, e + f$ values will be obtained.
Step 4: The efficiency of the suppliers should be determined using Eq. (10). Therefore, we can sort the suppliers based on $\theta_k^{DEA-OPA}$ values.

It should be noted that the type of research methodology based on the general classification and data collection is quantitative research, according to the nature

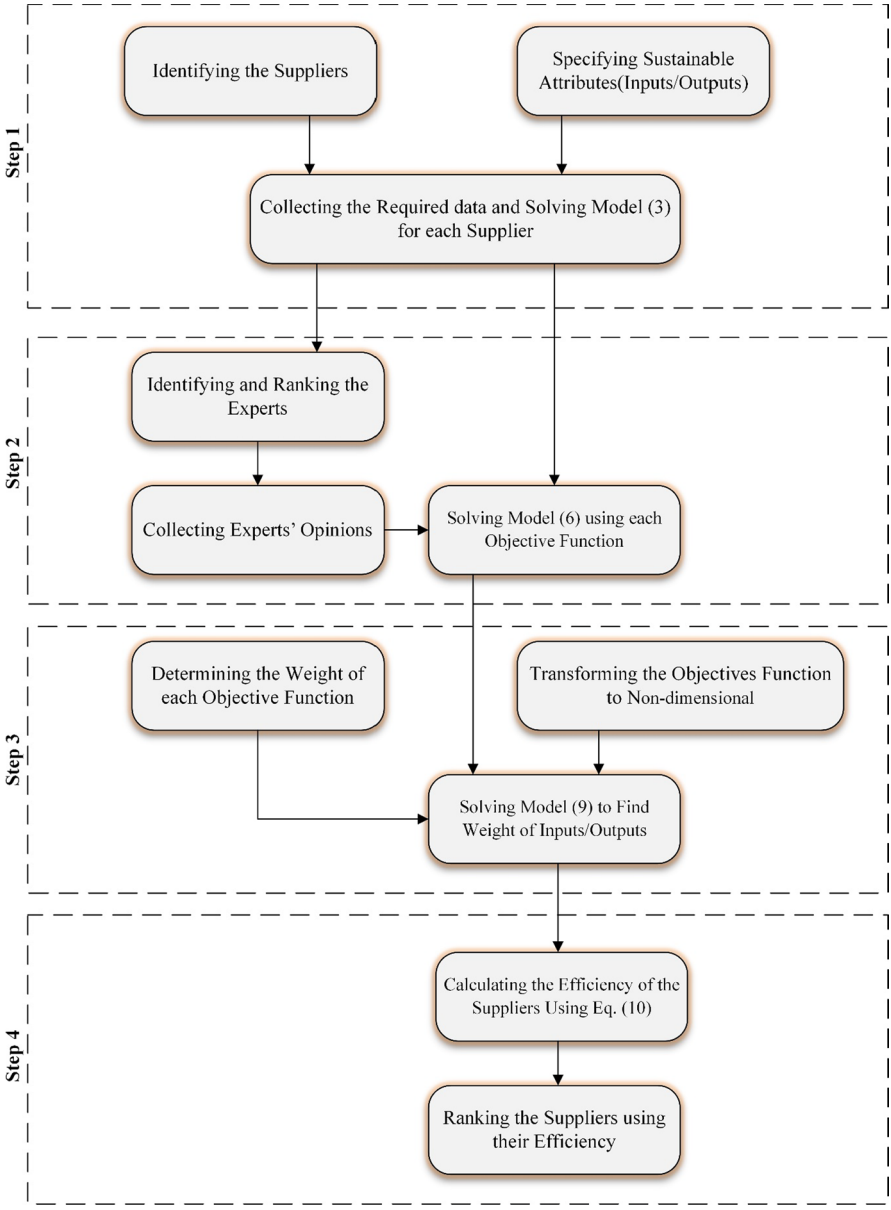

Fig. 1 The proposed framework for determining suppliers' efficiency

of the study is prescriptive, and according to the purpose of the study is applied research. For further clarification about the proposed framework, a numerical example has been presented in the next section, which can be useful to implement the proposed framework.

7 Numerical Example

In this section, we aim to address a numerical example to check the feasibility of the proposed framework under different situations for performance measurement of the suppliers. The steps are as follows (based on Fig. 1):

Step 1: The DEA data associated with the example has been presented in Table 2. There are six suppliers in this example, while the manager needs to determine suppliers' efficiency and select the best one for the company based on sustainability attributes.

Using the data in Table 2, Model (3) is constructed and solved for each supplier separately to find the value of θ_k^{CCR} $(k = 1, ..., m)$. The results are presented in Table 3.

Step 2: Three experts are identified. Also, it should be noted that the preference of the experts has been considered as Expert 1 > Expert 3 > Expert 2. As mentioned earlier, the preference of the experts can be defined based on their work

Table 2 The input/output data of the suppliers for the numerical example

DMUs	Inputs		Output	
	Environmental cost	Cost of production	Green product	On-Time delivery
Supplier 1	223	530	30	32
Supplier 2	260	310	63	50
Supplier 3	189	435	49	23
Supplier 4	215	370	23	15
Supplier 5	195	545	78	43
Supplier 6	301	396	46	28

Table 3 The value of θ_k^{CCR} for each supplier

Suppliers	θ_k^{CCR}
Supplier 1	0.6497953
Supplier 2	1.0000000
Supplier 3	0.7376619
Supplier 4	0.3634657
Supplier 5	1.0000000
Supplier 6	0.6006259

Table 4 The experts' opinions of the inputs/outputs attributes for the numerical example

Inputs/Outputs	Rank		
	Expert 1	Expert 2	Expert 3
Environmental cost	1	4	1
Cost of production	4	3	4
Green product	2	2	3
On-time delivery	3	1	2

Table 5 The value of objective functions of model (6) for the numerical example

Objective function	f_k^*	f_k^{min}
f_1	−0.367158	−151.402300
f_2	0.000000	−260.000000
f_3	−2.311296	−139.418100
f_4	−0.267089	−78.145130
f_5	0.000000	−195.000000
f_6	−2.532860	−180.788400
f_7	0.136364	0.000000

experience, knowledge, or other relevant factors. The experts' opinions regarding the inputs and outputs are presented in Table 4.

Using the data in Tables 2, 3, 4, Model (6) is constructed and solved separately by each objective function. The obtained results are presented in Table 5.

Step 3: It should be noted that the weights of DEA objectives and the OPA objective have been considered equally in this example ($S_{DEA}= S_{OPA} = 0.5$). Using the data in Tables 2, 3, 4, 5, Model (9) is constructed and solved. The weights of inputs/outputs are illustrated in Table 6. As can be seen from Table 6, "Environmental Cost" is the most important one, and "Cost of Production" has the least important degree among them.

Step 4: This step (Step 4 in Fig. 1) calculates the suppliers' efficiency using the obtained weights in Table 6. For this purpose, Eq. (10) has been utilized, and the results are provided in Table 7. Moreover, the ranks of suppliers are extracted based on the efficiency values.

Table 6 The weights of the inputs/outputs for the numerical example

DEA-OPA ($S_{DEA}= S_{OPA} = 0.5$)		
Inputs/outputs	Weight	Rank
Environmental cost	0.404101	1
Cost of production	0.071724	4
Green product	0.218670	3
On-time delivery	0.305505	2

Table 7 The efficiency of the suppliers for the numerical example

Suppliers	Efficiency	Rank
Supplier 1	0.127499	4
Supplier 2	0.228211	2
Supplier 3	0.164922	3
Supplier 4	0.084747	6
Supplier 5	0.256113	1
Supplier 6	0.124056	5

As can be seen from Table 7, Supplier 5 is the most efficient one among them. However, there is not a considerable difference between the efficiency of Suppliers 2 and 5. It means that Supplier 2 can be a good choice for the company as well.

7.1 Comparative Analysis

In this section, we intend to compare the results of the DEA-OPA approach with the DEA and OPA models individually. For this purpose, the previous numerical example has been solved by each model, and the results in terms of the suppliers' efficiency were presented in Table 8 and Fig. 2. It should be noted that "DEA (Supplier 1)" represents the situation in which the type of the model is DEA, and "Supplier 1" is functional DMU.

As can be seen from Fig. 2 and Table 8, the DEA-OPA line has been located between the DEA and OPA lines consistently. Indeed, the DEA-OPA is a synthetic model combining the DEA and the OPA models with the aid of multiple objective models, while it can make a balance between human judgment and the quantitative data of the suppliers. One of the key advantages of the DEA-OPA approach is to allow the decision-maker to specify the degree of importance for OPA and DEA (Weights of the objective function). Here, the same preference has been employed (S_{DEA}= S_{OPA} = 0.5), but it can be different based on the need. Moreover, the rank of the suppliers in each model has been provided in Table 9 and Fig. 3. Based on Table 9, in almost all situations in DEA and OPA results, suppliers 5 and 2 have the highest efficiency, respectively. Therefore, the same results should be obtained for the DEA-OPA model. This procedure can be utilized to check the results of the proposed approach for other suppliers, which shows that the results of DEA-OPA are reasonable.

As can be seen from Fig. 3, the result of the DEA-OPA is a combination of the DEA and OPA. In fact, the proposed model can balance the human judgment (OPA) and actual data (DEA) of the suppliers.

7.2 DEA-OPA under Incomplete Input Data

In this section, we aim to address the performance of the proposed approach, while the input data of the OPA is incomplete. For this purpose, Table 4 has been changed to Table 10.

In Table 10, expert 1 has no opinion regarding " Environmental Cost ". In fact, when the expert does not have sufficient information about inputs/outputs, the response sheet is incomplete. In this case, the constraints related to missing values should be removed from the OPA constraints in Model (9). After utilizing the steps in Fig. 4, the efficiency of the suppliers has been obtained in Table 11. Based on Table 11, the obtained ranks are different for Suppliers 1 and 6 compared with the complete case for the DEA-OPA model (see Table 9), but there is no change for other suppliers.

Table 8 The results of the DEA, OPA, and DEA-OPA models for the numerical example in terms of efficiency

DMUs	DEA (Supplier 1)	DEA (Supplier 2)	DEA (Supplier 3)	DEA (Supplier 4)	DEA (Supplier 5)	DEA (Supplier 6)	OPA	DEA-OPA
Supplier 1	0.673216	0.278527	0.374890	0.374890	0.336323	0.374890	0.108434	0.127499
Supplier 2	1.000000	1.000000	1.000000	1.000000	0.605769	1.000000	0.198227	0.228211
Supplier 3	0.574482	0.554279	0.737662	0.737662	0.648148	0.737662	0.148600	0.164922
Supplier 4	0.346049	0.305877	0.363466	0.363466	0.267442	0.363466	0.074651	0.084747
Supplier 5	1.000000	0.704238	1.000000	1.000000	1.000000	1.000000	0.228268	0.256113
Supplier 6	0.478323	0.571589	0.600626	0.600626	0.382060	0.600626	0.109809	0.124056

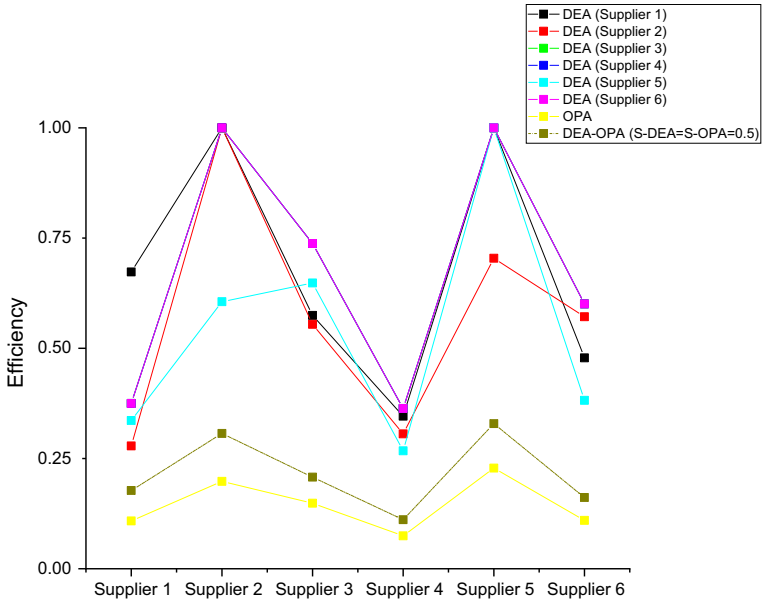

Fig. 2 The comparison among DEA, OPA, and DEA-OPA models in terms of efficiency

Table 9 The results of the DEA, OPA, and DEA-OPA models for the numerical example in terms of suppliers' rank

DMUs	DEA (Supplier 1)	DEA (Supplier 2)	DEA (Supplier 3)	DEA (Supplier 4)	DEA (Supplier 5)	DEA (Supplier 6)	OPA	DEA-OPA
Supplier 1	3	6	5	5	5	5	5	4
Supplier 2	2	1	1	2	3	2	2	2
Supplier 3	4	4	3	3	2	3	3	3
Supplier 4	6	5	6	6	6	6	6	6
Supplier 5	1	2	2	1	1	1	1	1
Supplier 6	5	3	4	4	4	4	4	5

For more investigation, Fig. 4 has been provided. Figure 4 shows the behavior of the DEA-OPA when there is a missing value in the OPA input data. As can be seen from Fig. 4, the OPA-DEA moves towards the DEA solution when the OPA is incomplete. In Fact, the DEA-OPA model tries to receive the information from DEA to handle the missing values in the OPA. Therefore, its line should be fallen between DEA-OPA (complete case) and DEA solutions. Thus, changing the rank of suppliers 1 and 6 in the case of missing value seems reasonable.

In the next section, a pilot application has been provided to show the application of the proposed model in a real-world situation. Also, a sensitivity analysis has

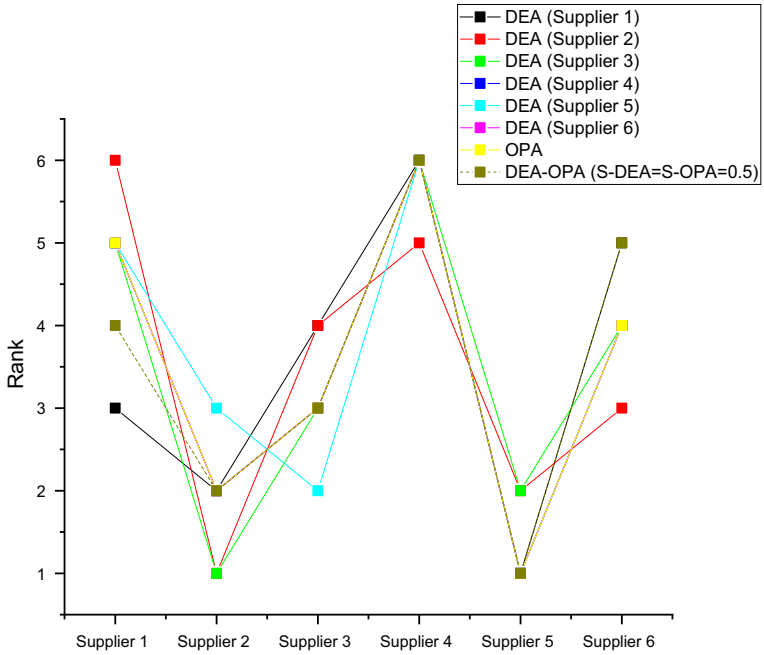

Fig. 3 The comparison among DEA, OPA, and DEA-OPA models in terms of suppliers' rank

Table 10 The experts' opinions of the inputs/outputs attributes for the numerical example with incomplete data

Inputs/Outputs	Rank		
	Expert 1	Expert 2	Expert 3
Environmental cost	–	4	1
Cost of production	3	3	4
Green product	1	2	3
On-time delivery	2	1	2

been presented to check the sensitivity of the proposed method for changing different factors.

8 A Pilot Application

Sustainable development is essential for humankind's long-term survival. Its philosophy is critically important for some industries that have a direct impact on the environment. For example, the paper products industry is one of them because of the enormous effects on the environment. Many companies in this industry have to present sustainable development reports every year. In the current study, some famous companies in the paper industry have been selected to determine their efficiency by

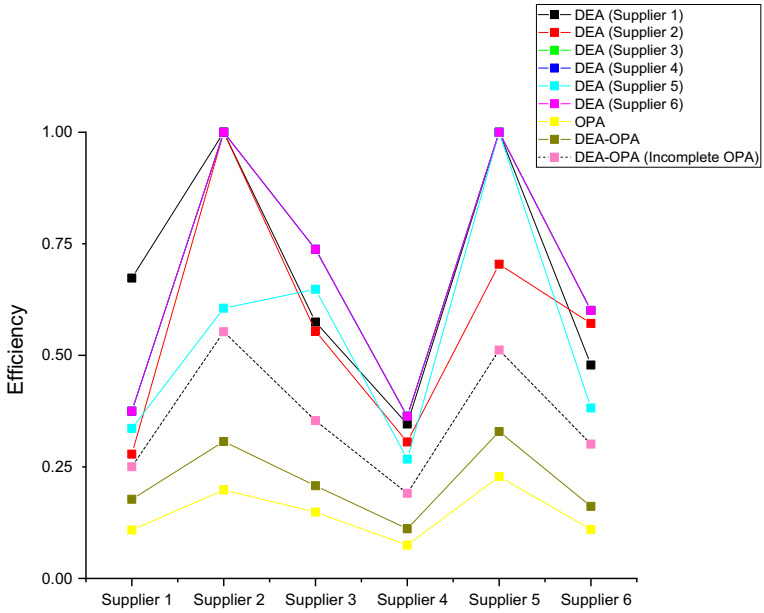

Fig. 4 The comparison of the results between the complete and incomplete data sets

Table 11 Comparison of the results of DEA-OPA (Complete case) with DEA-OPA (Incomplete case)

DMUs	DEA-OPA (Incomplete case)		DEA-OPA (Complete case)	
	Efficiency	Rank	Efficiency	Rank
Supplier 1	0.250311	5	0.127499	4
Supplier 2	0.552725	1	0.228211	2
Supplier 3	0.353601	3	0.164922	3
Supplier 4	0.190834	6	0.084747	6
Supplier 5	0.51196	2	0.256113	1
Supplier 6	0.301149	4	0.124056	5

Table 12 The input/output data of the suppliers for the pilot application

Suppliers	Inputs		Outputs		
	Manufacturing and product costs (Million$) (−)	Greenhouse Gas Emissions (thousand tons) (−)	Production (thousand tons) (+)	Net income (Million $) (+)	Women Employed (+)
Supplier 1	632.48	61.61	1120.00	22.57	1622.00
Supplier 2	1822.13	44.00	838.00	143.77	572.00
Supplier 3	5069.00	349.00	2500.00	283.00	2000.00
Supplier 4	1821.56	4468.93	6000.00	974.06	5411.00
Supplier 5	3377.00	43.00	4380.00	379.00	2500.00
Supplier 6	4913.24	1132.41	3000.00	314.74	8188.00

Table 13 The experts' opinions of the inputs/outputs attributes for the pilot application

Inputs/Outputs	Expert 1	Expert 2	Expert 3	Expert 4
Manufacturing and product costs (Million$) (−)	2	1	3	2
Greenhouse Gas Emissions (thousand tons) (−)	1	3	1	3
Production (thousand tons) (+)	3	2	4	1
Net income (Million $) (+)	4	5	5	4
Women Employed (+)	5	4	2	5

Table 14 The weights of the inputs/outputs for the pilot application

DEA-OPA (S_{DEA}= S_{OPA} = 0.5)		
Inputs/outputs	Weight	Rank
Manufacturing and product costs (Million$) (−)	0.460551	1
Greenhouse gas emissions (thousand tons) (−)	0.258571	2
Production (thousand tons) (+)	0.160233	3
Net income (Million $) (+)	0.04247	5
Women employed (+)	0.078175	4

the proposed method. A part of the data has been collected from the companies' annual report, and other parts have been collected from the experts. The quantitative information regarding the inputs/outputs of the suppliers have been presented in Table 12. To selecting the inputs/outputs, the experts' opinions, and availability of the real-world data have been considered. Also, three crucial dimensions of the sustainability approach have been covered. This question may arise why "Greenhouse Gas Emissions" is input or "Women Employed" is output? Since "Greenhouse Gas Emissions" is a negative attribute, it should be considered as input in DEA. Also, since "Women Employed" is a positive attribute, it should be regarded as output. All inputs should be negative, and all outputs should be positive for using the DEA model, which is essential for employing the DEA model.

On the other hand, the experts' opinions about the inputs/outputs have been provided in Table 13. Here, the problem is group decision-making, and there are four experts for the pilot application who had enough knowledge about the sustainability context.

It is worth mentioning that the experts have been ranked based on their working experience as follows:

Expert 3 = Expert 4 > Expert 1 > Expert 2.

After employing steps 1 to 3 in Fig. 1, the weights of the inputs/outputs have been obtained in Table 14. As can be seen from Table 14, "Manufacturing and product costs" is the most important one among them, and "Net income" has the lowest value among the group of inputs/outputs. The results seem entirely reasonable based on the input data of the OPA.

Table 15 The efficiency of the suppliers for the pilot application

DMUs	Efficiency	Rank
Supplier 1	1.000000	1
Supplier 2	0.217618	6
Supplier 3	0.234641	5
Supplier 4	0.714867	2
Supplier 5	0.583092	3
Supplier 6	0.443794	4

Based on Fig. 1, after calculating the weights of the inputs/outputs, the efficiency of the suppliers should be determined using Eq. (10). Finally, the results have been presented in Table 15.

As can be seen from Table 15, Supplier 1 has the highest efficiency score among the group of potential suppliers. Also, the results have been acknowledged by experts. After that, Supplier 4 achieved the second position with an efficiency value of 0.714867. It seems that Supplier 4 can be a good alternative when we need more than one supplier. Supplier 5 achieved the third position compared with other suppliers, yet it has not an excellent efficiency (0.583092), and selecting that is not suggested to the policymakers. In the next section, we intend to analyze the sensitive factors in the DEA-OPA model.

8.1 Sensitivity Analysis

Sensitivity analysis plays a vital role in legitimizing a decision-making process. A small change can affect the results seriously, and the reliability of the results should be checked before utilizing it. The DEA-OPA model enjoys a high level of flexibility because the decision-maker can specify which model (OPA or DEA) has a higher impact on the results. Here, three states have been considered includes "DEA > OPA", "DEA = OPA", and "DEA < OPA". It is interesting to mention that with the aid of the original OPA, we can determine the weights of objective functions in each state. For example, if we use the original OPA for the state "DEA > OPA", the weights of DEA and OPA will be 0.75 and 0.25, respectively. After utilizing the proposed steps in Fig. 1 for these three states, the results of the pilot application have been illustrated in Fig. 5.

As can be seen from Fig. 5, the states "DEA > OPA" and "DEA = OPA" have the same results, yet the result for the state "DEA < OPA" is different from the two mentioned states. In state "DEA < OPA", Supplier 4 achieved the second position and Supplier 5 the third position. These results show that determining the preference for the DEA and OPA can affect the results and should be considered carefully.

Another sensitive factor in the DEA-OPA model is the preference of the experts. Experts should be ranked based on their working experience, academic degree, etc. Here, we have provided two scenarios as follows, while the first

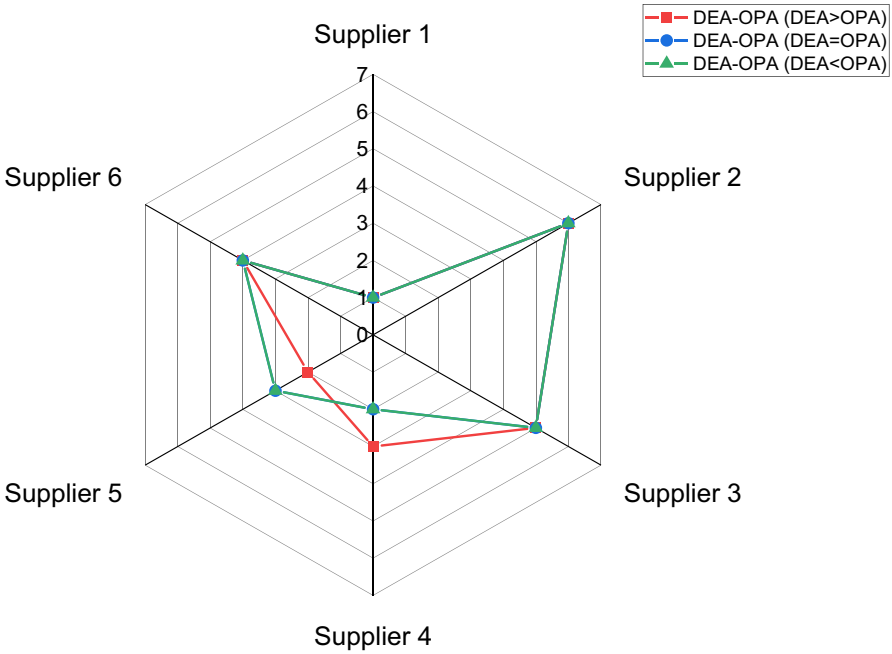

Fig. 5 The rank of suppliers for different preferences of DEA and OPA

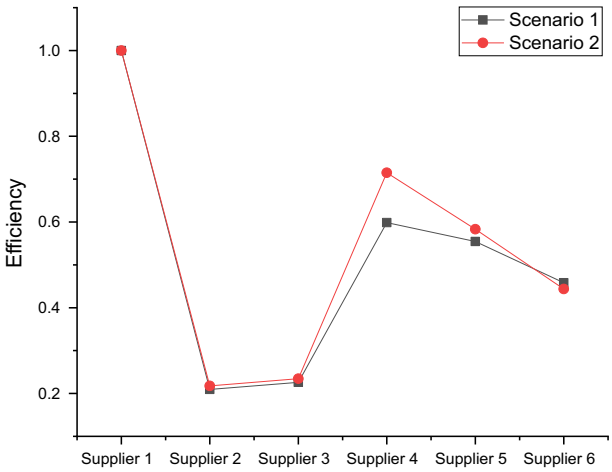

Fig. 6 The efficiency of suppliers for the different preferences of the experts

one is based on their academic degree, and the second one is based on working experience.

Scenario 1: Expert 1 = Expert 3 > Expert 2 > Expert 4.

Scenario 2: Expert 3 = Expert 4 > Expert 1 > Expert 2.

After solving the pilot application for these two scenarios, the results have been obtained in Fig. 6.

It should be noted that the ranks of suppliers have not experienced any change in this pilot application, but the efficiency of the suppliers has been changed. Notably, this change is a lot for suppliers 4 and 5. Therefore, this factor may change the results in some situations, and decision-maker should pay special attention to specifying them. These advantages of the DEA-OPA model make it flexible and realistic.

9 Discussion and Conclusion

DEA is one of the most useful methods to determine the efficiency of the suppliers, which has been employed by scholars several times. However, it can be extended to consider human judgments during the decision-making process, leading to more realistic solutions. A human can consider many factors in his/her mind, while it is too difficult to consider them by the original DEA model. Therefore, the current study has proposed the DEA-OPA model to overcome this shortcoming. The DEA-OPA model can consider the experts' opinions and quantitative data to calculate the efficiency of the suppliers. It is worth mentioning that when experts lack sufficient knowledge, there is no need to provide false information. Indeed, the DEA-OPA can handle incomplete data in case of a lack of information, which was addressed in numerical example 1.

Moreover, the preference of the experts can be considered during calculating the efficiency of the suppliers, which can lead to new results based on the sensitivity analysis section. Since the weighted Min–Max approach has been utilized to solve the multiple-objective model, the decision-maker can specify the weights of the DEA and the OPA objective functions as well. It is worth mentioning that the proposed approach not only can be used for sustainable supplier evaluation problems but also for other types of supplier evaluation problems by changing the inputs/outputs. The feasibility of the model has been checked in different situations through a numerical example and a pilot application. Also, the results have been compared with the original DEA and the OPA models. The pilot application has been selected from the paper industry, considering its direct impact on the environment.

One of the most critical limitations of the current study is collecting data for some important factors, such as the number of injuries in the companies. The medical incident rate (MIR) can be extracted based on the sustainability reports, yet we could not simultaneously employ the ratio with other data types in the DEA model. Therefore, several limitations in terms of selecting the right form of input and outputs for the pilot application need addressing in the future. Further, the proposed study can just handle the problems with incomplete data in experts' opinions. In some situations, the input data of the DEA is incomplete as well. Therefore, the scholars can propose a new DEA-OPA model that can handle the incomplete data in both DEA and OPA models simultaneously. Moreover, a new model based on fuzzy set theory can be proposed, which can be employed in uncertain situations.

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Declarations

Conflict of interest No potential conflict of interest was reported by the authors.

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